Consider a binary linear code $\mathcal{C}(n,k)$ with minimum distance decoding:

Prove that, for every $\underline{c} \in \mathcal{C}$

$$D(\underline{c}) = \underline{c} + D(\underline{0})$$

Take $D(\underline{c}) = \{\underline{y} \in \mathbb{F}_2^n : d_H(\underline{y},\underline{c}) \leq d_H(\underline{y},\underline{c}'), \forall \underline{c}' \in \mathcal{C}\}$

Therefore

$$D(\underline{c}) = \underline{c} + D(\underline{0}) = \underline{c} + \{\underline{y} \in \mathbb{F}_2^n : d_H(\underline{y},\underline{0}) \leq d_H(\underline{y},\underline{c}'), \forall \underline{c}' \in \mathcal{C}\}$$

We can see $d_H(\underline{y},\underline{0})$, but the distance between the vector $\underline{0}$ and $\underline{y}$ is the number of 1 in $\underline{y}$, so

$$d_H(\underline{y},\underline{0}) = w_H(\underline{y})$$

$$D(\underline{0}) = \{\underline{y} \in \mathbb{F}_2^n : w_H(\underline{y}) \leq d_H(\underline{y},\underline{c}'), \forall \underline{c}' \in \mathcal{C}\}$$

We can write $\underline{y}$ as $\underline{y} = \underline{c} + \underline{e}$, where $\underline{e}$ is the difference between $\underline{c}$ and $\underline{y}$, so

$$D(\underline{0}) = \{(\underline{c}+\underline{e}) \in \mathbb{F}_2^n : w_H((\underline{c}+\underline{e}) \leq d_H(\underline{c}+\underline{e},\underline{c}'), \forall \underline{c}' \in \mathcal{C}\}$$

$$\exists \underline{e}\left(w_H(\underline{c}+\underline{e}) \leq d_H(\underline{c}+\underline{e},\underline{c}'), \forall \underline{c}' \in \mathcal{C}\right), \underline{c},\underline{e} \in \mathbb{F}_2^n \Rightarrow$$

$$\Rightarrow D(\underline{c}) = \underline{c} + D(\underline{0}) \Rightarrow D(\underline{0}) = D(\underline{c}) + \underline{c} \Rightarrow$$

$$\Rightarrow D(\underline{0}) = \{\underline{y} \in \mathbb{F}_2^n : d_H(\underline{y}, \underline{c}) \leq d_H(\underline{y}, \underline{c}'), \forall \underline{c}' \in C\} + C$$

We can see $D(\underline{c}) = \{\underbrace{\underline{y}_0, \underline{y}_1, \dots, \underline{y}_{N-1}}_{N}\}$

Where $\underline{y}_i = (\underline{c} + \underline{e}_i)$, $i = 0, \dots, N-1$

Substituting,

$$D(\underline{c}) = \{(\underline{c} + \underline{e}_0), \dots, (\underline{c} + \underline{e}_{N-1})\}$$

Where write $D(\underline{0})$ as

$$D(\underline{0}) = \{\underline{c} + \underline{e}_0, \dots, \underline{c} + \underline{e}_{N-1}\} + C = \{\underline{e}_0, \underline{e}_1, \dots, \underline{e}_{N-1}\}$$

So $D(0)$ is the set containing every difference vector from $\underline{c}$ where the $w_H(\underline{e}_i) \leq d_H(\underline{e}_i, \underline{c}') \; \forall \underline{c}' \in C \wedge i = 0, \dots, N-1$

We can rewrite it as

$$D(\underline{0}) = \{\underline{e} \in \mathbb{F}_2^n : w_H(\underline{e}) \leq d_H(\underline{e}, \underline{c}'), \forall \underline{c}' \in C\}$$

Therefore

$$\underline{e} \in D(\underline{0}) \wedge \underline{c} \in \{\underline{c}\}$$

$$\Rightarrow \forall \underline{c} \; \exists \underline{e} \left(\underline{c} + \underline{e} \in (D(\underline{0}) \cup \{\underline{c}\})\right), \underline{c} \in \{\underline{c}\} \wedge \underline{e} \in D(0)$$

we defined $\underline{c} + \underline{e} = \underline{y}$ before, therefore

$$\underline{y} \in (D(\underline{0}) \cup \{\underline{c}\}) \wedge \underline{y} \in D(\underline{c}) \wedge D(\underline{c}) = D(\underline{0}) \cup \{\underline{c}\} \Rightarrow$$

$$\Rightarrow D(\underline{c}) = \underline{c} + D(\underline{0})$$



$\underline{c}' = \underline{c} + \underline{c}'' \Rightarrow c'' = \underline{c} + \underline{c}'$ and $\underline{y} = \underline{c} + \underline{e}$

$$D(\underline{c}) = \{\underline{y} \in \mathbb{F}_2^n : d_H(\underline{y}, \underline{c}) \leq d_H(\underline{y}, c'), \forall c' \in C\} =$$

$$= \{\underline{y} \in \mathbb{F}_2^n : d_H(\underline{y}+\underline{c}, \underline{c}+\underline{c}) \leq d_H(\underline{y}+\underline{c}, \underline{c}'+\underline{c}), \forall c' \in C\}$$

$$= \{\underline{y} \in \mathbb{F}_2^n : d_H(\underline{e}, \underline{0}) \leq d_H(\underline{e}, c''), \forall c' \in C\} =$$

$$= \{\underline{y} \in \mathbb{F}_2^n : w_H(\underline{e}) \leq d_H(\underline{e}, \underline{c}''), \forall c' \in C\}$$

$\underline{c}'' \hookrightarrow \underline{c} + \underline{c}'$

Now we take

$$D(\underline{0}) = \{\underline{y} \in \mathbb{F}_2^n : d_H(\underline{y}, \underline{0}) \leq d_H(\underline{y}, c'), \forall c' \in C\} =$$

$$= \{\underline{y} \in \mathbb{F}_2^n : w_H(\underline{y}) \leq d_H(\underline{y}, \underline{c}'), \forall c' \in C\} =$$

$$= \{\underline{0}+\underline{e} \in \mathbb{F}_2^n : w_H(\underline{0}+\underline{e}) \leq d_H(\underline{0}+\underline{e}, \underline{c}'), \forall c' \in C\} =$$

$$= \{\underline{e} \in \mathbb{F}_2^n : w_H(\underline{e}) \leq d_H(\underline{e}, \underline{c}'), \forall c' \in C\}$$

In $D(\underline{c})$ we have $w_H(\underline{e}) \leq d_H(\underline{e}, \underline{c}'')$, while in $D(\underline{0})$ we have $w_H(\underline{e}) \leq d_H(\underline{e}, \underline{c}')$, knowing that the difference between $\underline{c}'$ an $\underline{c}''$ is $\underline{c}$

We can say that

$$D(\underline{c}) = D(\underline{0}) + \underline{c}$$

■